

PT LAPI GANESHA UTAMA

In technical partnership with the Institute of Technology Bandung

TECHNICAL CERTIFICATION DOCUMENT

IECEX/ATEX Certification Document**Gas Leak Detector (GLD) V2**

Prepared in direct reference to the IECEX/ATEX Certification Information Requirements issued by the certification body (ExCB) — covering Section 2, Technical Documentation (Items 1–6: product description; name, model, and specification list; functional description and technical parameters; product photographs; intended use and installation environment; and design and manufacturing information) and Section 3, Sample Information. Section 1, Basic Information (Application and Organization), remains to be completed in a subsequent revision.

Document Control

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Manufacturer	PT LAPI Ganesha Utama
Technical partner	Institute of Technology Bandung
Reference checklist	IECEX/ATEX Certification Information Requirements

Revision History

REVISION	DATE	DESCRIPTION
0.1	11 September 2026	Initial issue — Section 2, Items 1–6 (product description; name, model, and specification list; functional and technical parameters; product photographs; intended use and installation environment; design and manufacturing information) and Section 3 (sample information).

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About This Document

This document is a working technical file prepared to satisfy Section “2. Technical Documentation” and Section “3. Sample Information” of the official IECEx/ATEX Certification Information Requirements checklist issued by the certification body (ExCB). The table of contents follows the complete structure of the original checklist (Sections 1–3); Section 1 (Basic Information) remains pending and will follow in a subsequent revision.

Original excerpt, Section 2 & 3 — source: IECEx ATEX Certification Information Requirements (ExCB):

“1) Detailed product description; 2) Product name, model, and specification list; 3) Complete and clear functional description and technical parameters (electrical parameters, mechanical parameters, etc.); 4) Clear product photos (overall and key components); 5) Description of intended use and installation environment (e.g., gas group IIC/IIB/IIA if applicable, temperature group T1–T6, ambient temperature range, area classification 0/1/2 or 20/21/22); 6) Design and manufacturing information [a–i: drawings, BOM, material datasheets, manufacturing process, explosion-protection calculations, temperature group calculation, usage and installation instructions, nameplate information, Ex component certificates]. 3. Sample Information: 1) model, serial number, and status of the sample; 2) necessary test fixtures or auxiliary equipment.”

1. Basic Information (Application and Organization)

Covers the application form, business license/company registration, organizational chart and contact information, manufacturing facility address and profile, and (where applicable) ISO 9001 certification.

Not yet prepared in this revision. To be completed in a subsequent revision.

2. Technical Documentation

Six items per the original checklist (1–6, with Item 6 comprising sub-items a–i). This revision addresses the complete section: Items 1–6.

2.1 • Detailed Product Description

The Gas Leak Detector (GLD) is an IoT-based, multi-sensor gas leak detection device designed for the early detection of flammable and process gases in oil & gas refinery environments (process units, tank farms, pipe racks, and storage/loading-unloading areas). This document addresses the GLD unit itself as the subject of the current IECEx/ATEX certification.

Functionally, the GLD integrates eight channels of metal-oxide semiconductor gas sensors (MQ series), an edge-AI microcontroller/processor (ESP32-S3), and a LoRa radio module (star-topology wireless transmission) within a single fixed-point unit installed at locations with gas-leak risk. An on-device AI gas-classification model runs directly on the unit so that detection decisions do not depend on a continuous connection to a central server. When gas concentration exceeds a defined threshold, the unit triggers a local alarm (an integrated visual/audible alarm module) and simultaneously transmits an alarm notification over the LoRa network to the operator dashboard.

The enclosure is designed for hazardous-area deployment at refinery sites, using metal materials (aluminum alloy and stainless steel — no plastic or PVC) and mounted via an L-bracket to existing structures without drilling or

welding. Important: this design-intent statement does not constitute a claim that the enclosure has passed testing or has been Ex-certified — the explosion-protection scheme, gas group, temperature class, and target installation zone will be addressed in Section 2.5 (to follow).

The current production power configuration is continuous 24 VDC, supplied via an AC/DC adapter connected to the site electrical supply. A portable battery power path (Li-ion 18650) remains under development (R&D) and has not become a deployed production configuration.

2.2 · Product Name, Model, and Specification List

Product name	Gas Leak Detector (GLD) — Node Sensor
Model / version	GLD V2 (Version 2)
Manufacturer	PT LAPI Ganesha Utama
Technical development partner	Institute of Technology Bandung — IoT Laboratory & Physics Laboratory
End client / program owner	PT Pertamina Patra Niaga (initial deployment site: Refinery Unit IV, Cilacap)
Primary function	Acquisition of 8-channel gas sensor data and LoRa transmission
Microcontroller	ESP32-S3-WROOM-1U-N16R8
LoRa radio module	E22-900MM22S
Dimensions (L×W×H)	200 × 90 × 290 mm
Enclosure material	Aluminum alloy + stainless steel

Source: official product technical datasheet (Institute of Technology Bandung, Revision 4.0), cross-referenced with internal technical specification documentation and EMC parameter measurement data.

2.3 · Functional Description and Technical Parameters (Electrical, Mechanical, etc.)

Functional workflow (normal operating mode): sense → process → transmit. Each of the eight gas sensors continuously samples ambient conditions → data is normalized and classified by the on-device AI model (ESP32-S3) → the result is transmitted over the LoRa network (star-topology transmission) at a configurable interval (default 10 seconds), or immediately (event-driven) when an alarm condition is detected. Gas alarms are triggered through two parallel channels: a local visual/audible alarm module on the unit itself, and a push notification transmitted over the LoRa network to the dashboard — the alarm-push pathway has been successfully tested on a campus mesh network (field validation at a production refinery installation is still pending).

2.3.a · Electrical Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Main power input	24 VDC	Final — production config.	Continuous power supply from a site AC/DC adapter.
AC/DC adapter input	220 VAC, 50 Hz	Final	Compliant with the Indonesian national electrical grid standard (PLN).

AC/DC adapter output	24 VDC	Final	
Internal operating voltage	5 VDC & 3.3 VDC	Final	5 VDC for the MQ sensor/heater circuitry; 3.3 VDC for ESP32-S3 logic. Per-rail current not yet documented separately.
Maximum input current	≈0.33 A @ 24 VDC	Calculated	Calculated from measured maximum power consumption (7.995 W) divided by 24 VDC.
Maximum power consumption	7.995 W @ 24 VDC	Measured — production config.	Applies to the continuous-power configuration. The battery (R&D) configuration is recorded separately at 5.75 W — a different operating mode, not a data conflict.
Backup battery path (R&D, not in production)	Li-ion 18650 cells, 7 in parallel, 4.2 V/cell, ≈28,000 mAh total	Development pathway	Not yet a deployed production configuration. Cell/BMS safety certification (e.g., UN 38.3, IEC 62133) not yet obtained. Firmware-reported diagnostic thresholds: low battery at 3.50 V, critical at 3.30 V (status/flag only, not an active power cutoff).
Electrical protection (fuse, reverse polarity, overvoltage, overcurrent)	—	Pending confirmation	Scope of electrical protection not yet defined/documented.

2.3.b · Sensor and Communication Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Gas sensors	MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135 (8 channels)	Final	Metal-oxide semiconductor sensors; sensing element directly exposed to ambient air.
AI gas-classification model	On-device classifier — 3 classes: Clean Air, LPG, H ₂	Final	Runs locally on the ESP32-S3 (Running/Inference mode); outputs a class label and a confidence value. Does not yet cover CO ₂ , Benzene, CO, or H ₂ S.
Environmental sensor (temperature/humidity)	SHT40-AD1B-R2 (I2C)	Final	Auxiliary temperature/humidity input included in the primary hardware design; values are available for status/telemetry.
Analog-to-digital converter	ADS1256IDBR	Final	24-bit multi-channel analog acquisition for the 8 sensor channels; 30,000 SPS, firmware SPI clock 1.92 MHz.
Processing unit	ESP32-S3-WROOM-1U-	Final	Certified under FCC (2AC7Z-ESPS3WROOM1U), TELEC, and CE

	N16R8		(per Espressif data) — RF/EMC certifications, not an “Ex component” certification.
Communication module	LoRa, E22-900MM22S module	Final	Certified under CE, FCC, and RoHS (per Ebyte data) — RF/EMC certifications, not an “Ex component” certification.
Operating frequency	920 MHz	Final	Star-topology transmission; within the regional 920–923 MHz ISM band (Indonesia).
Transmit power (firmware configuration)	17 dBm	Final	The radio module supports up to 22 dBm — 17 dBm is an operational configuration, not the module's maximum limit.
Bandwidth / spreading factor / coding rate	125 kHz / SF7 / CR 4/5	Final	Source: EMC parameter table and official product technical datasheet.
Antenna	External, omnidirectional, SMA male connector, 3 dBi gain	Final	On some units, the 2.4 GHz Wi-Fi antenna remains inside the enclosure and must be relocated externally.
Data transmission interval	Configurable, default 10 seconds	Final	Alarm events are transmitted immediately, independent of the periodic interval.
Other interfaces	SPI, LoRa	Final	External ports/connectors: USB, sensor, power, fan, antenna, alarm buzzer.
RS-485 / Modbus interface	Read-only Modbus RTU slave, 9600 bit/s 8N1, Unit ID 1 (THVD1410DR transceiver)	Final	Provides 8 read-only registers (device status word, classification result, confidence, battery voltage, power source, external power, LoRa transmission counter, node ID); not used for product control.

2.3.c • Mechanical Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Enclosure material	Aluminum alloy + stainless steel	Specific grade pending	PVC is not used in any housing or bracket component. Specific grade still under determination.
Dimensions (L × W × H)	200 × 90 × 290 mm	Final	Consistent between the technical specification documentation and the EMC parameter table.
Total weight	—	Pending confirmation	Not yet weighed/documented.
Mounting method	L-bracket, following the design	Final	Mounted to existing structures

	already installed at the refinery		without drilling or welding.
Ingress protection (IP rating)	—	Pending confirmation	Not yet tested/determined.
Cable entry (gland)	—	Pending confirmation	Cable gland specification not yet determined.
Antenna mounting	External, SMA male connector	Final	
Operating temperature	—	Pending confirmation	Ambient operating temperature range not yet determined — key parameter for temperature class (T1–T6) determination in Section 2.5.
Operating humidity	—	Pending confirmation	

Source: internal technical specification documentation, Sections 1.1–1.3 (Node Sensor), cross-referenced with the official product technical datasheet (Revision 4.0) and component certification verification (ESP32-S3-WROOM-1U, E22-900MM22S).

Rows marked “Pending confirmation” above do not reflect a documentation oversight — this is an honest status indicator. These fields are intentionally left blank because no official data yet exists (not yet measured, tested, or decided). They must not be filled with estimates in future revisions without a clear supporting data source.

2.4 • Product Photographs — Overall and Key Components

The seven photographs below were taken directly from the Node Sensor (GLD) V2 prototype unit (no capture-date metadata is available in the source files). The first three photographs show the fully assembled unit from different angles; the remaining four show key components in close-up.



Complete unit — front view

The fully assembled Node Sensor (GLD): the main enclosure (blue) with sensor mesh cover, external LoRa antenna, and a visual alarm module (cylindrical housing, red lens) mounted on the right side.



Complete unit — alternate angle

An alternate view of the assembled unit, showing the position of the antenna, alarm module, and the cable-gland side of the enclosure.



Complete unit — front view (without antenna and alarm module)

The main enclosure with the sensor mesh cover installed, shown in a flat front view illustrating the enclosure shape and its four mounting bolt holes.



Key component — PCB inside enclosure (before sensor modules)

The main PCB installed inside the enclosure prior to gas sensor module installation. The ESP32-S3-WROOM-1U microcontroller, LoRa radio module, and power circuitry are visible.



Key component — PCB with eight gas sensor modules

The same PCB with eight channels of MQ-series gas sensors (MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135) installed around the central microcontroller.



Key component — visual alarm module (close-up)

The visual alarm (beacon/strobe) module, constructed of metal with a red lens, connected to the main enclosure via a dedicated cable gland.



Key component — sensor mesh cover (close-up)

The metal mesh cover that protects the gas sensing elements while serving as a diffusion path for gas to reach the sensors beneath.

Photographic completeness — status as-is. The seven photographs above cover the complete unit and its key components (PCB, sensor modules, alarm module, mesh cover), substantively satisfying checklist Item 2.4. Still outstanding: (a) formally labeled photographs of each face (front/back/left/right/top/bottom) with a scale reference, as is customary in ExCB submission packages; (b) separate photographs of individual components such as the battery, gaskets/seals, terminals, and cable glands.

2.5 • Description of Intended Use and Installation Environment (Gas Group, Temperature Class, Area Classification)

The GLD is intended for continuous, fixed-point monitoring of flammable and toxic gas leaks in oil & gas refinery hazardous areas — process units, tank farms, pipe racks, and loading/unloading areas. The specific Ex classification parameters below reflect the project engineering team's current recommendation, developed from the intended deployment envelope and the gases the device is designed to detect; none of these parameters has yet been confirmed with the certification body (ExCB) or a notified body, and none should be treated as final.

PARAMETER	RECOMMENDED VALUE	STATUS	RATIONALE / REMARKS
Gas group	IIC	Team recommendation — pending ExCB confirmation	Hydrogen (H ₂) is one of the three gases the classifier is designed to detect (Section 2.3.b); IIC is required for hydrogen and inherently covers IIB and IIA.
Temperature class	T4 (≤135°C)	Team recommendation — pending verification	Not yet substantiated by measurement. MQ-series metal-oxide sensors rely on an internal heating element as their normal operating principle; a hot-spot measurement specific to the sensor models and enclosure configuration used here has not yet been performed (see 2.6.f).
Ambient temperature range	—	Pending confirmation	Not yet measured or specified.
Area classification (zone)	Zone 1	Team recommendation — pending ExCB confirmation	Assessed as sufficient for the general refinery deployment envelope, including areas near storage. Zone 0 would only apply if the detector were installed directly inside a tank vapor space.

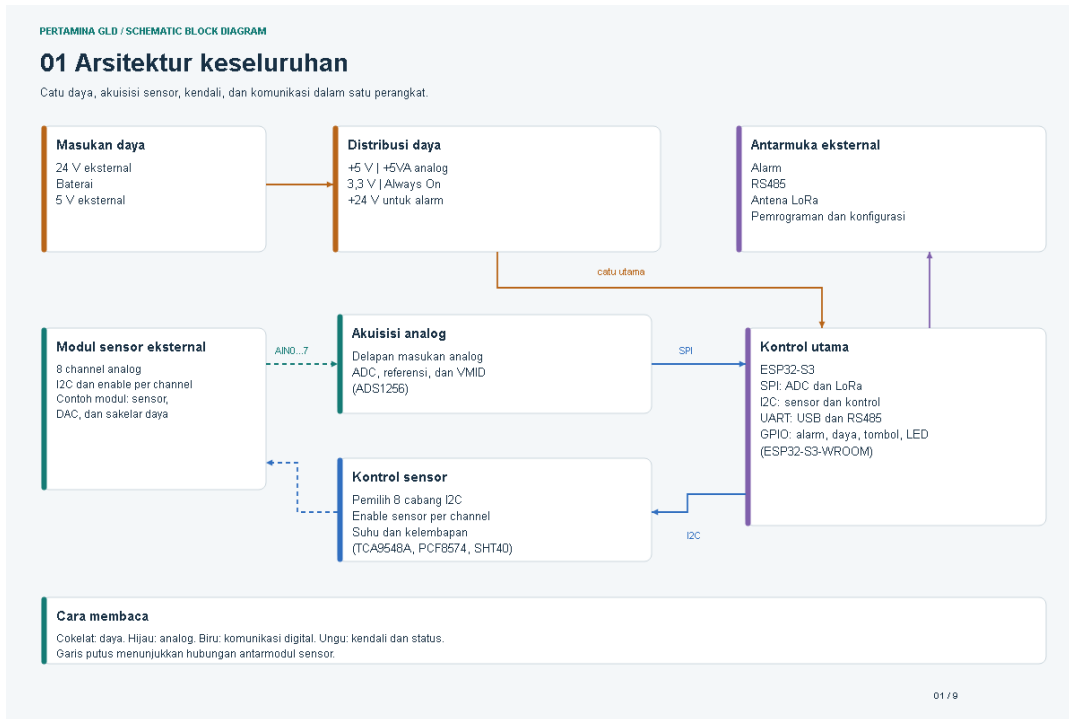
These are engineering recommendations, not a certification decision. An independent readiness assessment of the same checklist item, prepared separately from the underlying firmware/hardware repository, reached the same conclusion — gas group, temperature class, ambient range, and zone are not yet formally established. Final classification requires explicit agreement with the ExCB.

2.6 • Design and Manufacturing Information

Nine sub-items (a–i) per the original checklist. Status is reported item by item below; most sub-items are not yet available — this is reported plainly rather than implied to be complete.

2.6.a · Complete Drawings (Assembly, Component, Electrical Schematic, PCB Layout, Enclosure Structure, Junction Box, Terminal, Grounding)

An electrical schematic capture and a corresponding PCB layout exist for the GLD V2 main board (EDA source design files). From the schematic's traced net list, a supporting block-diagram set (9 sheets, functional/block level, 204 components mapped with documented pin-to-net traceability) has been produced and is reproduced in full below. Field labels in the source diagrams are in Indonesian; English captions are provided under each sheet.



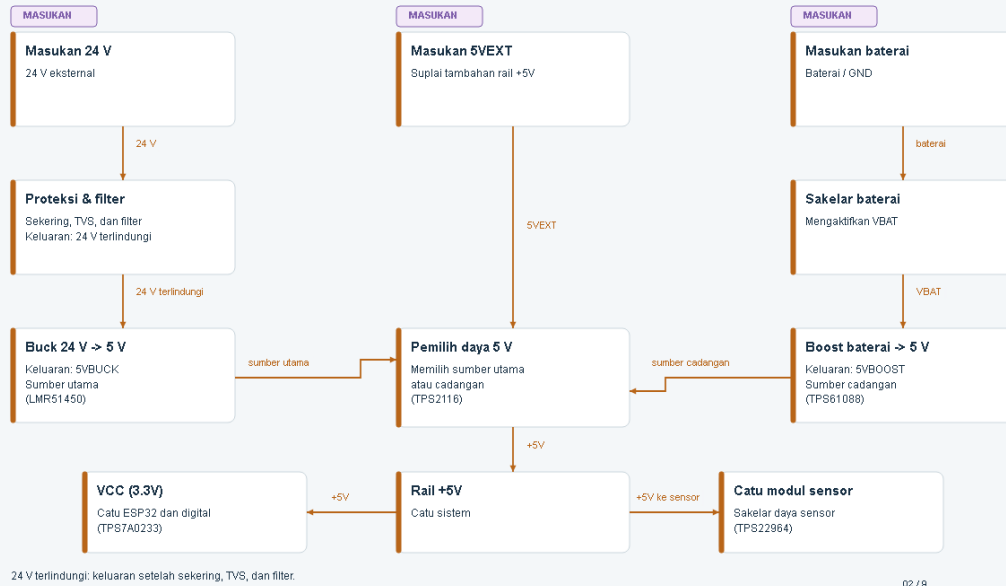
Sheet 1 of 9 — Overall architecture

Power input, power distribution, analog acquisition, sensor control, main control (ESP32-S3), and external interfaces (alarm, RS-485, LoRa antenna, programming/configuration).

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

02 Distribusi daya utama

Tiga masukan daya, pemilihan sumber 5 V, rail analog, dan catu sensor.

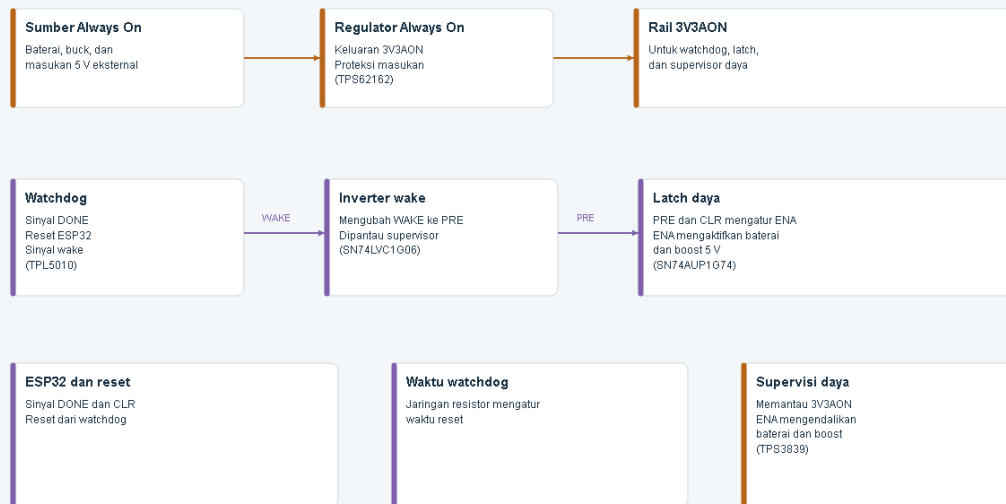
**Sheet 2 of 9 — Main power**

Main power input rails and distribution: +5 V / +5VA analog, 3.3 V, and +24 V for the alarm circuit.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

03 Always On, watchdog, dan latch

Always On (AON) adalah rail daya yang tetap aktif untuk menjaga pengawasan dan pengendalian daya.

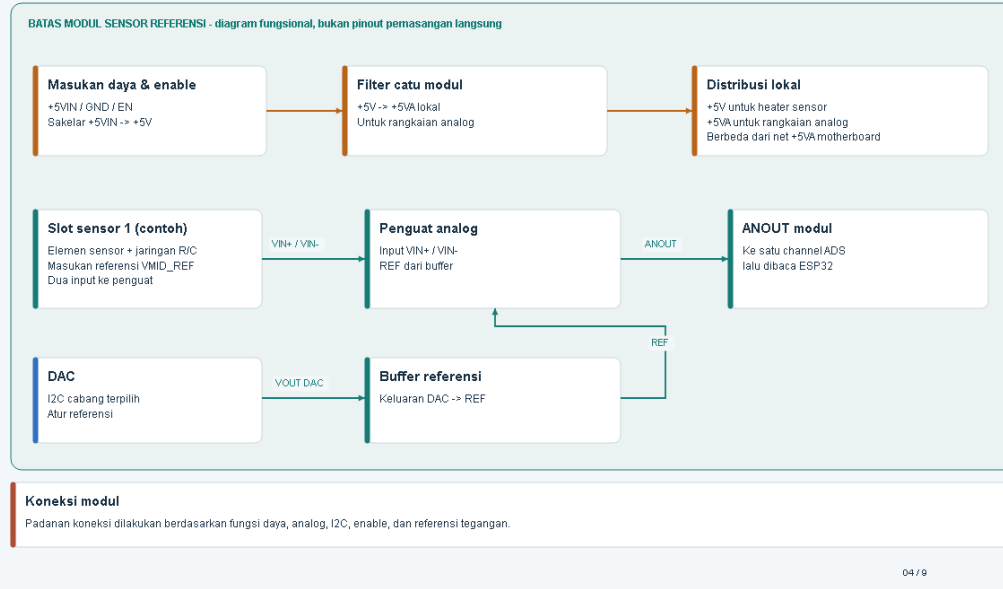
**Sheet 3 of 9 — Always-on power**

The always-on power domain that remains active independent of the main control state.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

04 Modul sensor: jalur fungsional

Setiap modul menerima daya, I2C, enable, dan menyediakan satu keluaran analog.



Sheet 4 of 9 — Sensor module

The external gas sensor module: 8 analog channels with per-channel I2C enable, including the sensor, DAC, and power-switch sub-blocks.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

05 Akuisisi analog dan tegangan referensi

Jalur analog memiliki filter catu terpisah, referensi tegangan, VMID, dan ADS 8 Channel.



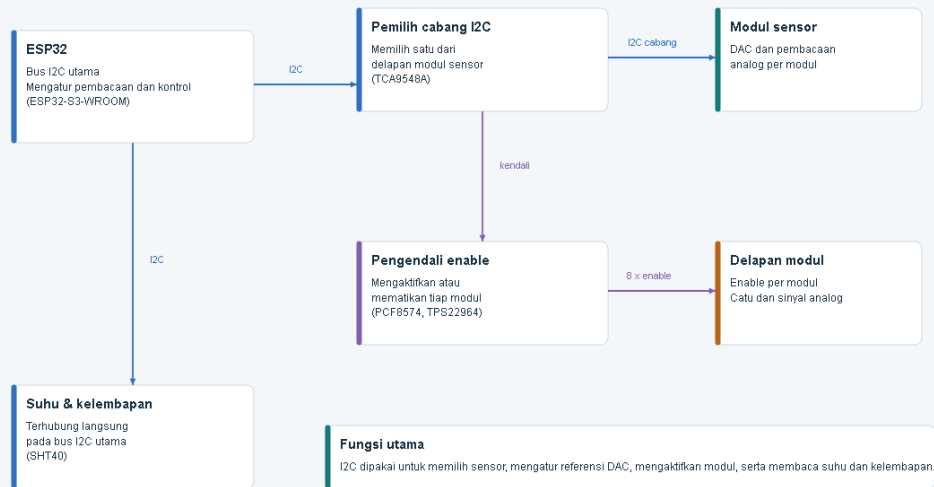
Sheet 5 of 9 — Analog acquisition

The 8-channel analog acquisition front end: analog inputs, ADC (ADS1256), reference, and VMID biasing.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

06 I2C dan kendali sensor

I2C memilih cabang sensor, mengatur DAC, dan mengendalikan daya tiap modul.



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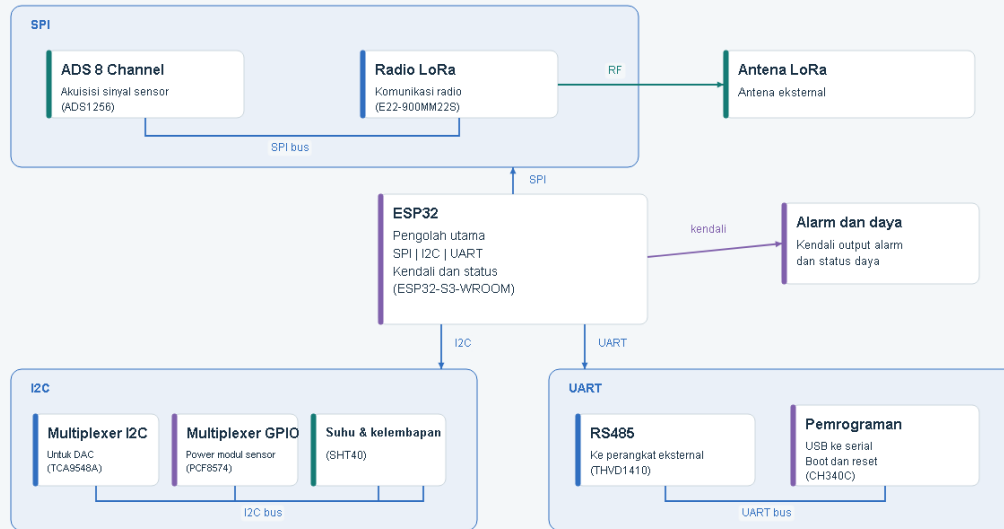
Sheet 6 of 9 — I2C & sensor control

I2C channel selection (TCA9548A), per-channel sensor enable (PCF8574), and the temperature/humidity sensor (SHT40).

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

07 Komunikasi dan koneksi ESP32

Koneksi dikelompokkan menurut bus SPI, I2C, dan UART, kendali perangkat dipisahkan dari komunikasi.



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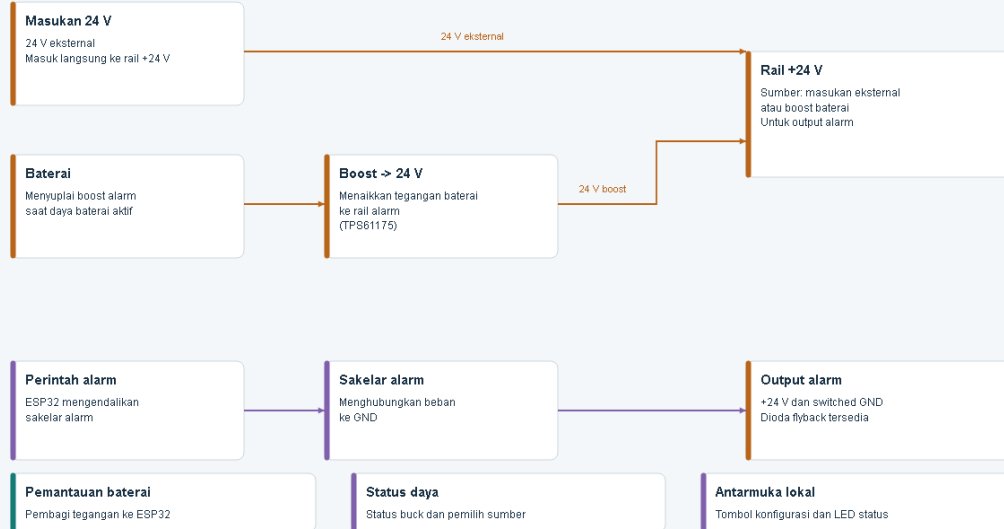
Sheet 7 of 9 — ESP32 connections

Main controller (ESP32-S3) interface map: SPI to the ADC, I2C to sensor/control, UART for USB and RS-485, and GPIO for alarm, power, button, and LED.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

08 Alarm, baterai, dan status daya

Rail +24 V alarm dapat berasal dari masukan 24 V eksternal atau boost baterai.



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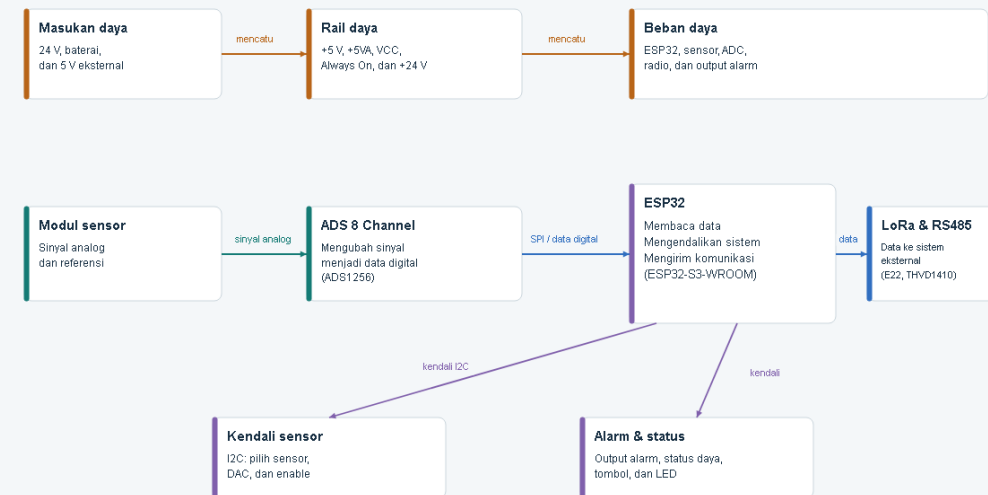
Sheet 8 of 9 — Alarm & status

Alarm output and status-indication circuitry driven from the main controller.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

09 Alur energi, data, dan kendali

Daya mencatu perangkat; data sensor mengalir ke ESP32; ESP32 mengendalikan komunikasi dan keluaran.



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Sheet 9 of 9 — System flow

Overall signal and data flow across the board, summarizing how the preceding eight sheets connect end to end.

DRAWING TYPE	STATUS	REMARKS
Electrical schematic (component-level) / block diagram	Partially available	Schematic capture and a derived 9-sheet block-diagram set exist with traceability evidence (pin-

		to-net mapping). Not yet issued in a released, revision-controlled drawing format with a formal drawing number.
PCB layout	Partially available	PCB layout export exists for the same board revision as the schematic above.
Assembly drawing	Not yet available	
Component drawing	Not yet available	
Enclosure structure drawing (gap, length, volume)	Not yet available	Required if a flameproof (Ex d) protection concept is pursued.
Junction box / terminal / grounding connection drawings	Not yet available	

The schematic and PCB layout are useful supporting engineering artifacts, but on their own they do not constitute the certified drawing package the checklist requires — dimensioned, toleranced, material-annotated drawings in a released revision-controlled form have not yet been produced.

2.6.b • Bill of Materials (BOM) for Explosion-Safety-Relevant Components

A controlled, Ex-critical BOM (enclosure, gaskets, terminals, cable entry devices, switches, light sources, battery, potting compound, plastic parts, printed circuit boards — with manufacturer, model, material grade, and certification/technical parameters for each) has not yet been compiled. A component-level pin/net export exists from the schematic (204 components traced) and can serve as a starting reference, but manufacturer part numbers, material grades, and Ex/UL/CCC certification status for the safety-critical items above have not been determined. Comparative research on Ex-rated enclosure products from other manufacturers exists internally as a reference for target specifications only — it describes third-party products, not this product's actual components, and is not included here.

Status: Not yet available.

2.6.c • Material Specification Sheets / Datasheets (Non-Metallic Materials)

Datasheets or supplier conformity declarations for non-metallic materials (enclosure components, seals, insulators, potting compounds) — covering heat/cold resistance, anti-aging, anti-static, flame retardancy, CTI value, and chemical resistance — have not yet been collected.

Status: Not yet available.

2.6.d • Manufacturing Process Description

A description of manufacturing processes relevant to explosion-protection safety (enclosure machining accuracy control, explosion-proof surface treatment, welding, potting, die-casting, bonding) has not yet been documented.

Status: Not yet available.

2.6.e · Explosion-Protection Calculations and Explanations (if applicable)

Calculations depend on the explosion-protection concept selected (e.g., Ex d, Ex e, Ex i), which has not yet been confirmed with the ExCB. No calculations have been performed.

Status: Not yet available — pending protection-concept confirmation.

2.6.f · Temperature Group Calculation (Hottest-Point Temperature Estimation)

No hottest-point temperature calculation or measurement has been performed for this product. As general context: MQ-series metal-oxide gas sensors operate using an internal heating element, a class of sensor commonly associated with published operating temperatures in the approximate 200–400°C range — however, this is a general characteristic of the sensor class, not a measured value for the specific sensor models, drive circuitry, and enclosure configuration used in this product. A worst-case hot-spot measurement is identified as the top-priority action required to substantiate the recommended T4 classification in Section 2.5.

Status: Not yet available.

2.6.g · Usage and Installation Instructions (Draft)

SUB-ITEM	STATUS	REMARKS
a) Safety warnings	Not yet available	No Ex-specific safety warnings have been drafted.
b) Installation requirements (cable entry, torque, grounding, cleaning)	Partially available	A mechanical mounting method exists separately (L-bracket, installed to existing structures without drilling or welding — see Section 2.3.c), but cable-entry method, torque values, grounding requirements, and cleaning requirements have not yet been formally specified.
c) Operating instructions and maintenance requirements	Partially available	A firmware command/operation reference exists (Serial, MQTT, and LoRa command protocol for engineering use), but it covers software operation and commissioning — not Ex-specific maintenance or inspection requirements (frequency, content, precautions).

Status: Partially available — not yet consolidated into an Ex-specific installation and operation manual for ExCB review.

2.6.h · Nameplate Information

Nameplate artwork cannot yet be finalized: it depends on the certificate number, protection marking, temperature class, ambient range, IP rating, and serialization scheme — none of which has been assigned yet.

Status: Not yet available.

2.6.i • Ex Component Certificates

No components in this design currently hold an Ex component certificate. As noted in Section 2.3.b, the processing unit (ESP32-S3-WROOM-1U-N16R8) and the communication module (E22-900MM22S) hold RF/EMC certifications (FCC, TELEC, CE, RoHS) — these are not Ex component certificates and do not satisfy this item.

Status: Not yet available.

3. Sample Information

Model, serial number, and sample status (whether the unit can be powered on and operated), together with any required test fixtures or auxiliary equipment.

3.1 • Model, Serial Number, and Status

No formal sample register or dossier for ExCB submission has been established. Internal engineering and bench testing reference the “GLD V2” board configuration under firmware environment gld_v2; a discrete unit serial-numbering scheme for certification samples has not yet been implemented.

Status: Not yet available.

3.2 • Test Fixtures and Auxiliary Equipment

A draft internal functional test plan identifies the minimum equipment anticipated for bench-level verification (this plan has not yet been executed and is not evidence of lab readiness):

TEST GROUP	MINIMUM EQUIPMENT
Firmware / serial	Engineering commissioning tool or serial terminal; firmware package with recorded version
I2C / ADC / DAC	No additional equipment for protocol-level acknowledgement/readback; multimeter or oscilloscope where physical voltage must be substantiated
Power / watchdog timer	Controlled 24 V supply, an applicable battery source, multimeter, and oscilloscope/logic analyzer
Alarm	The actual alarm load, multimeter/oscilloscope, and hearing protection if an audible buzzer is fitted
LoRa	At least one counterpart Cluster Head/Gateway device with recorded configuration
RS-485 / Modbus	An RS-485/USB-RS485 master with proper termination and the agreed register map

Status: Draft plan only — not yet executed, and not evidence of ExCB/laboratory test readiness.

This is a working document, prepared in stages, drafted in direct reference to the IECEx/ATEX Certification Information Requirements (original English/Mandarin version issued by the certification body). Data sources: the official product technical datasheet (Institute of Technology Bandung, Revision 4.0), internal technical specification documentation, EMC parameter measurement data, and product photography. Fields marked “Pending confirmation” are not yet final and must not be relied upon for procurement or certification purposes without further verification.